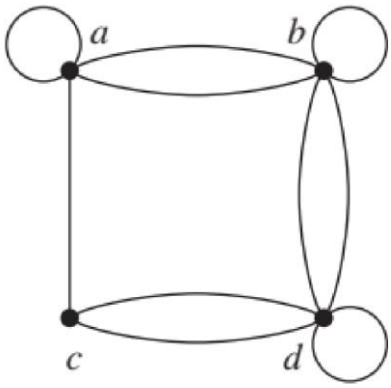
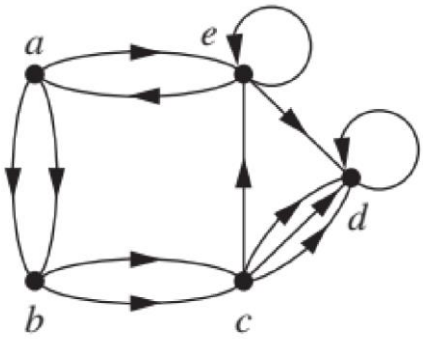
# Tutorial 4: Graph Theory I

Question 1: For each undirected graph that is not simple, find a set of edges to remove to make it simple.

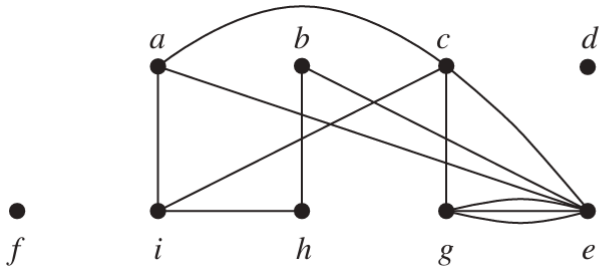
a)




b)



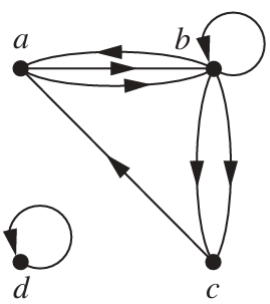

Question 2: Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.



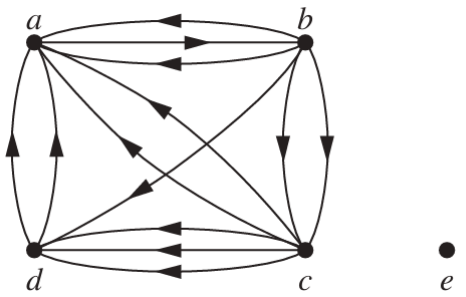

Question 3: Can a simple graph exist with 15 vertices each of degree five?

Question 4: Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

a)




b)

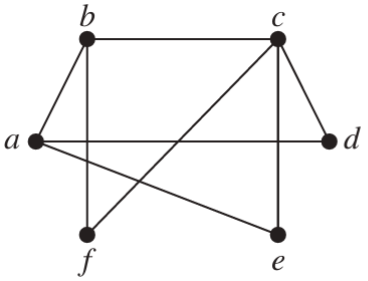



Question 5: Show that in a simple graph with at least two vertices there must be two vertices that have the same degree.

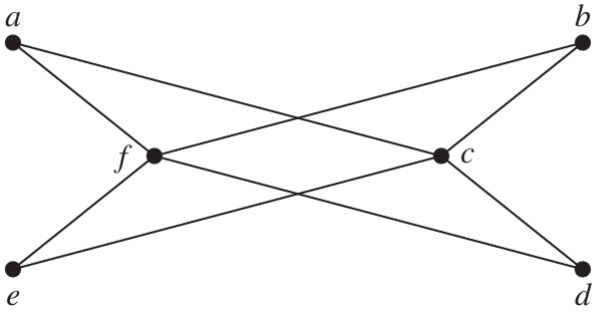
Question 6: Draw these graphs.

a) $K_7$
b) $C_7$
c) $K_{4,4}$
d) $W_7$
e) $Q_4$

Question 7: Determine whether the graph is bipartite.




a)




b)

Question 8: For which values of n are these graphs bipartite?

a) $K_n$

b) $C_n$

c) $W_n$

d) $Q_n$

Question 9: Let n be a positive integer. Show that a subgraph induced by a nonempty subset of the vertex set of $K_n$ is a complete graph.

Question 10: How many vertices and how many edges do these graphs have?

a) $K_n$
b) $K_{m,n}$
c) $C_n$
d) $Q_n$
e) $W_n$